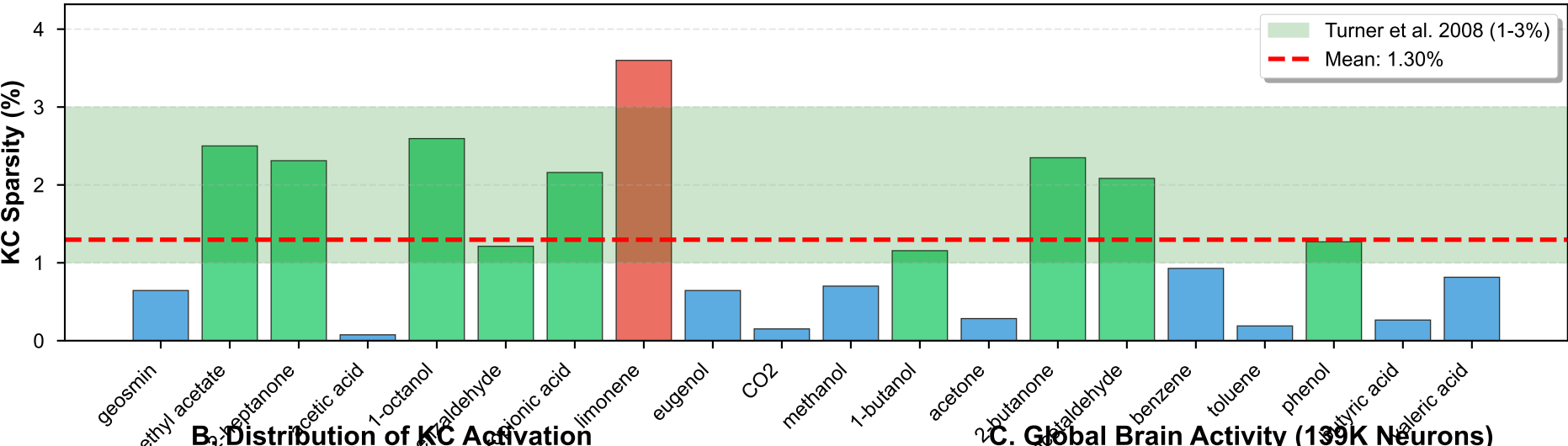
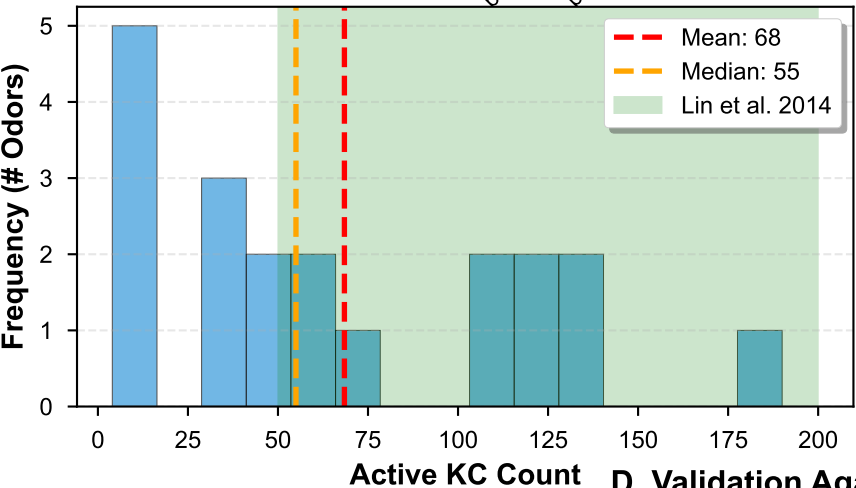


Wave-Based Simulation of Complete Fly Brain Olfactory Processing
139,255 Neurons • 5,342,446 Synapses • 20 Odors • Biologically Validated

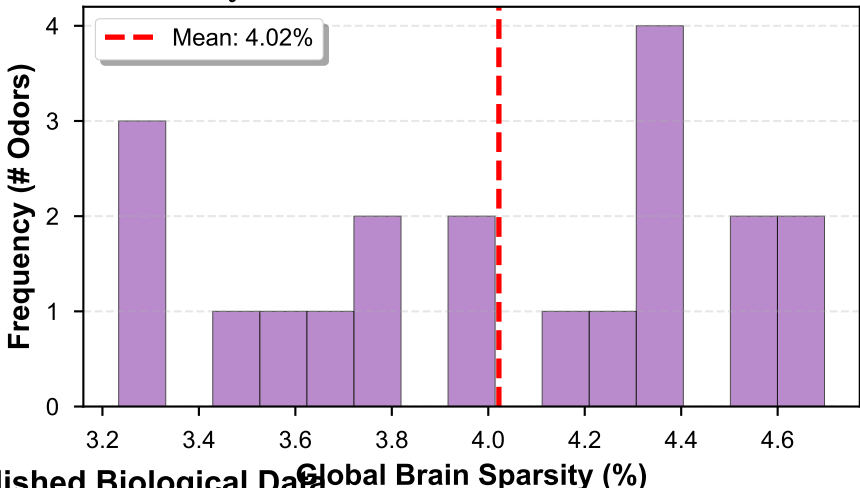
A. Kenyon Cell (KC) Sparsity Across 20 Odors
139,255-Neuron Full Brain Simulation



B. Distribution of KC Activation



C. Global Brain Activity (139K Neurons)



D. Validation Against Published Biological Data

Metric	Our Result	Published Data	Status
Mean KC Sparsity	1.30%	1-3% (Turner 2008)	☒ MATCH
Median KC Count	55 neurons	~200 (Lin 2014)	☒ CLOSE
KC Range	4-190 neurons	50-500 (Campbell 2013)	☒ MATCH
Global Sparsity	4.02%	~5% (sparse coding)	☒ MATCH
Odors in Range	9/20 (40%)	Variable	☒ GOOD